INCH-POUND

MIL-PRF-27/195A
18 July 2008
SUPERSEDING
MIL-T-27/195
18 October 1977

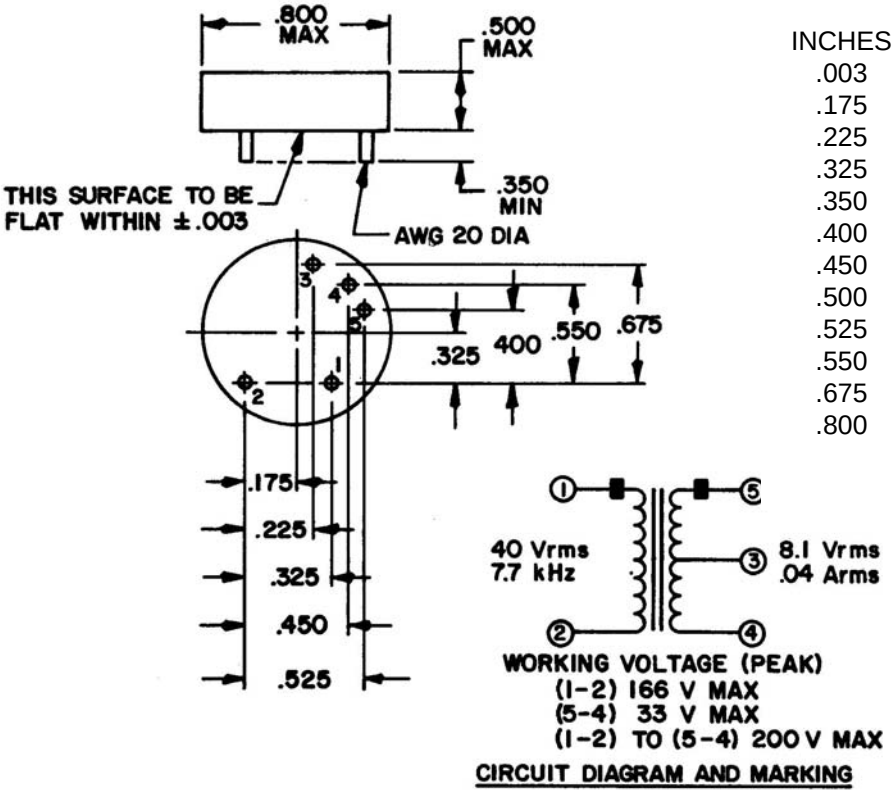
PERFORMANCE SPECIFICATION SHEET

TRANSFORMER, POWER, STEPDOWN

Inactive for new design
after 14 February 1997

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the products described herein shall consist of this specification and MIL-PRF-27.



INCHES	MM
.003	0.08
.175	4.44
.225	5.72
.325	8.26
.350	8.89
.400	10.16
.450	11.43
.500	12.70
.525	13.94
.550	19.97
.675	17.14
.800	20.32

NOTES:
1. Dimensions are in inches.
2. Metric equivalents are given for general information only.
3. Unless otherwise specified, tolerance is ±.005 (.13 mm).
4. Marking shall be on the top of the case.

FIGURE 1. Dimensions and configurations.

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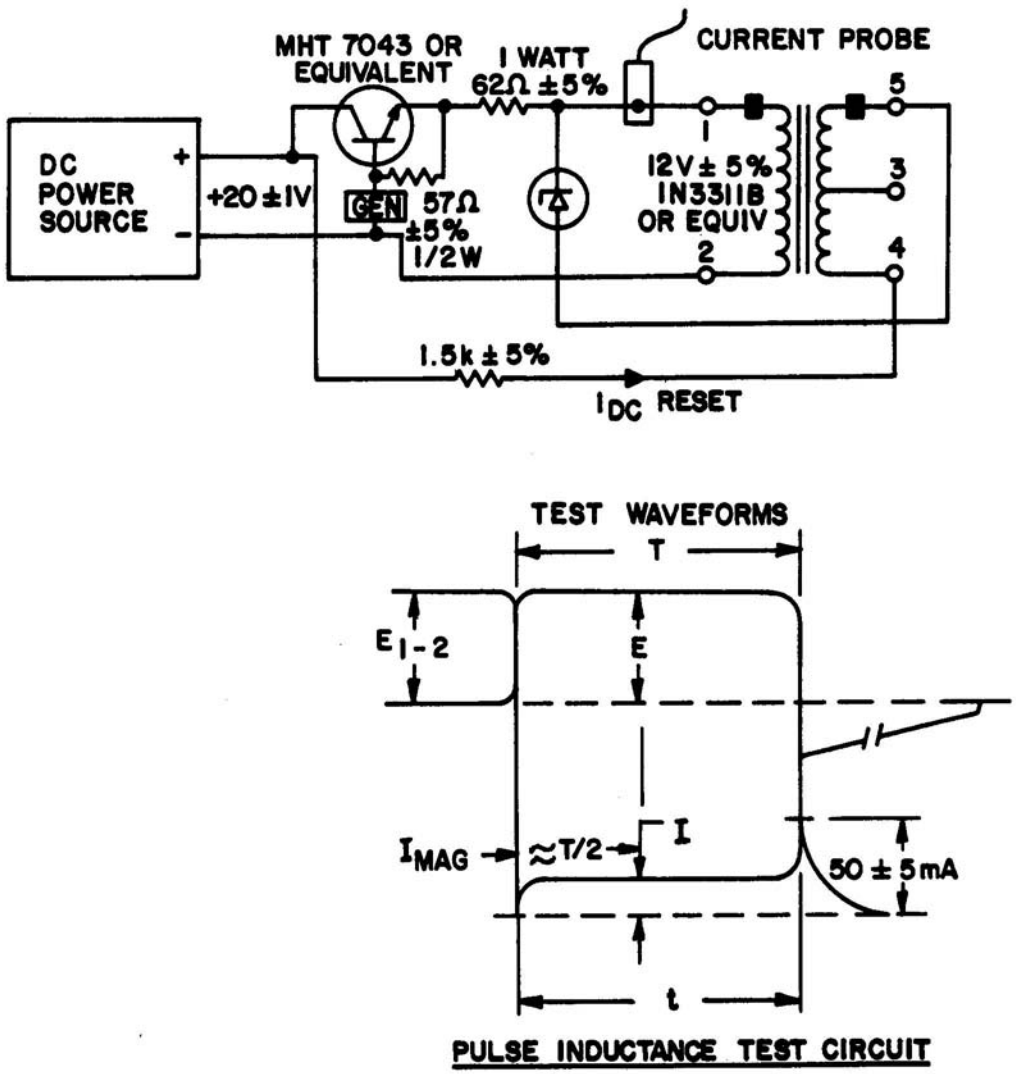


FIGURE 2. Pulse inductance test circuit.

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REQUIREMENTS: (When designations or colors in parentheses, i.e., (1-2) are used, they indicate the winding and the extreme terminals of the winding.)

Electrical ratings:

Primary voltage (1-2): 4 volts rms, 7.7 kilohertz.
Primary current: 4.0 milliamperes, rms.
Primary power: .15 volt ampere.
Secondary voltage (5-4): 8.1 volts rms, .04 ampere rms.
DC resistance:
Primary (1-2): 35 ohms, maximum.
Secondary (5-4): 2.15 ohms, maximum.
Working voltage:
(1-2): 166 volts peak, maximum.
(5-4): 33 volts peak, maximum.
(1-2) to (4-5): 200 volts peak, maximum.

Design and construction:

Dimensions and configuration: See figure 1.
Duty cycle: Continuous.
Case: Encapsulated.
Material: Epoxy.
Weight: 3 ounces.
Terminals: Pin type.
Material: Type S, tin, lead or lead alloy coating, in accordance with A-A-59551.
Height: .350 inch, minimum.
Diameter: AWG 20 (.032 inch).

Operating temperature range: -55ºC to +130ºC.

Terminal strength: MIL-STD-202, method 211, test condition A, 2 pounds.

Dielectric withstanding voltage (at sea level): 900 volts rms.

Electrical characteristics:

No load: With 40 volts rms and 7.7 kilohertz in (1-2):
Current in (1-2): 4.0 milliamperes rms.
Power in (1-2): .15 watt, maximum.
Voltage across (5-4): 8.08 volts rms ±1 percent.

Rated load: With 40 volts rms and 7.7 kilohertz (1-2):
Voltage across (5-4): 8.1 volts rms ±10 percent.

Pulse inductance: The transformer shall meet the following electrical requirements when tested under the following conditions using the test circuit in figure 2. The values shown shall be applicable to parts with a squareness ratio of 0.80 to 0.94.
a. Volt-millisecond product (ExT) = 2.85 ±5 percent at Imag – 50 milliamperes ±5. (For parts with a squareness ratio above 0.94, the ExT = 2.50 ±5 percent at Imag – 50 milliamperes ±5).
b. Linear current (I) = 3.4 milliamperes ±0.6 at time T/2. (For parts with a squareness ratio above 0.94, I = 7.0 milliamperes, maximum at time T/2.)
c. I Reset = 13.3 milliamperes ±10 percent.
d. Pulse width (T) = 240 microseconds, nominal. (For parts with a squareness ration above 0.94, the T = 200 microseconds, nominal.)

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e. Pulse rate – 200 hertz maximum.
f. Pulse voltage (E_{1-2}) = 12 volts ±5 percent.
g. Rise time (input pulse) = 0.5 microsecond, maximum.
h. Current probe: Tektronix type 6019 with model 134 amplifier or equivalent.

Polarity: Additive with terminals 2 and 5 connected.

Temperature rise: 20°C with 40 volts rms, 7.7 kilohertz across (1-2) at an ambient temperature of 125ºC.

Vibration (high frequency): MIL-STD-202, method 204.

Shock (specified pulse): MIL-STD-202, method 213, test condition A.

VERIFICATION:

Qualification inspection: Not applicable for this specification sheet.

Quality conformance inspection: Groups A and B tests of MIL-PRF-27 shall be applicable.

Marking location: See figure 1.

Part or Identifying Number (PIN): M27/195-01.

Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

Referenced documents. In addition to MIL-PRF-27, this document references the following:

MIL-STD-202
A-A-59551

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA - CC

Preparing activity:
DLA – CC

(Project 5950-2008-029)

Review activities:
Army – AR, CR4, MI
Navy – AS, MC, OS, SH
Air Force – 19, 99

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of this information above using the ASSIST Online database at http://assist.daps.dla.mil.

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